

Daily Organizer — Test Report

Field	Value
Date executed	2026-06-04
Environment	Local dev — backend <code>http://localhost:3000/api</code> , Firebase Realtime DB
Methodology	End-to-end API tests via PowerShell (<code>Invoke-RestMethod</code>) against the running backend, plus targeted source-code review
Test runner	test-runner.ps1 at the project root — re-runnable any time
Total tests written	60
Modules covered	Auth, Categories, Tasks, Goals + Milestones + Mini-goals, Habits, Journal, Projects + Payments, Focus (Pomodoro), Analytics, Dashboard, Cross-user isolation
Outcome	60 / 60 PASSING after 2 bug fixes

1. Summary

	Count
Tests passed (first run)	55
Tests failed (first run)	5
↳ True bugs in app code	2
↳ Cascading failures from those 2 bugs	3
↳ False positives in test runner	0 (after correcting refresh-token sequencing)
Tests passing after fixes	60 / 60
Files changed during testing	4 backend service files + 1 test runner

2. How to re-run

```
# Make sure backend is running on :3000
cd backend; npm run start:dev # terminal 1

# Then in a separate terminal:
cd ..
& '.\test-runner.ps1' # outputs PASS/FAIL per step + summary line
```

The runner registers a fresh test user (random email each run), seeds dummy data across every module, exercises happy + edge paths, then registers a second user to verify cross-user isolation. It's safe to re-run — every run creates new isolated records.

3. Full list of test cases

Each row is one assertion.  = passing in the latest run.

3.1 Auth (8 / 8)

#	Test	Status
1	Register new user (valid)	✓
2	Register with duplicate email returns 409	✓
3	Register with short password (< 8 chars) returns 400	✓
4	Login with correct credentials returns access + refresh tokens	✓
5	Login with wrong password returns 401	✓
6	GET /auth/me without token returns 401	✓
7	GET /auth/me with valid token returns the user profile	✓
8	POST /auth/refresh with a fresh refresh token issues new tokens	✓

3.2 Categories (4 / 4)

#	Test	Status
9	Create category with {name, color} only (no icon)	✓ (after BUG-001 fix)
10	Create category without name returns 400	✓
11	List categories returns the newly created one	✓
12	Update category (PATCH /categories/:id) changes the name	✓

3.3 Tasks / My Plans (8 / 8)

#	Test	Status
13	Create task (type='task', priority HIGH, with category + times)	✓
14	Create trip task with endDate (multi-day)	✓
15	Create task without title returns 400	✓
16	List tasks returns >= 2 records	✓
17	Get task by id matches	✓
18	Update task status='DONE' auto-stamps completedAt	✓
19	Filter tasks by ?status=DONE returns only DONE tasks	✓
20	GET /tasks/today returns an array (may be empty)	✓

3.4 Goals + Milestones + Mini-goals (4 / 4)

#	Test	Status
21	Create goal with description + resources, progress defaults to 0	✓
22	Add milestone to goal returns goal with milestones array	✓
23	Complete milestone bumps goal progress to 100 (1-of-1)	✓

24	Add mini-goal under milestone	✓
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3.5 Habits (7 / 7)

#	Test	Status
25	Create habit with weekdays [1,2,3,4,5] + time interval	✓
26	Create habit with invalid weekday (8) returns 400	✓
27	Create habit with bad startTime ("25:99") returns 400	✓
28	List habits with ?today=YYYY-MM-DD returns enriched fields (streak, history of 30, doneToday)	✓
29	Toggle today check-in flips doneToday true and increments streak	✓
30	Toggle today check-in again removes the check-in	✓
31	Creating a trip plan covering today makes isOffToday: true	✓

3.6 Journal (6 / 6)

#	Test	Status
32	PUT /journal/<today> upserts a new entry	✓
33	PUT with empty body (no body field) returns 400	✓
34	PUT same date again updates the existing entry (idempotent)	✓
35	GET /journal/<date> returns the saved entry	✓
36	GET /journal/<unknown date> returns null	✓
37	PUT with invalid date format ("not-a-date") returns 404	✓

3.7 Projects + Payments (7 / 7)

#	Test	Status
38	Create client project (isSelf=false, paymentStatus=PENDING)	✓
39	Create self project clears client/payment fields server-side	✓
40	Record payment on client project sets totalReceived and balance	✓
41	Recording payment on self project returns 403	✓
42	Update status to DELIVERED auto-stamps deliveredAt	✓
43	Update with partial body does not crash with "undefined" Firebase error (regression)	✓
44	Flipping isSelf=true clears client + payment fields	✓

3.8 Focus / Pomodoro (7 / 7)

#	Test	Status
45	Capture original task trackedMins (baseline for next test)	✓
46	Record WORK session with taskId	✓

47	WORK session auto-increments task's trackedMins by actualMinutes	✓
48	Record SHORT_BREAK session without task	✓
49	Invalid session type returns 400	✓
50	Negative actualMinutes returns 400	✓
51	GET /focus/today?today=... returns workSessions/workMinutes/breakMinutes/sessions	✓

3.9 Analytics (3 / 3)

#	Test	Status
52	GET /analytics/summary?range=30d returns full payload (tasks/habits/focus/journal/projects + 30-day daily + 7-day weekday + topCategories + insights)	✓
53	range=7d returns 7-element dailyActivity array	✓
54	range=junk returns 400	✓

3.10 Dashboard (2 / 2)

#	Test	Status
55	GET /dashboard/stats returns totalTasks etc.	✓
56	GET /dashboard/calendar?year=Y&month=M returns calendar data	✓

3.11 Cross-user isolation (4 / 4)

#	Test	Status
57	Register a second user (B)	✓
58	User B GET on user A's task → 403	✓
59	User B DELETE on user A's project → 403	✓
60	User B's task list does NOT contain any of user A's tasks	✓

4. Bugs found and fixed

BUG-001 — Categories: 500 error when creating without an icon

Field	Value
Severity	High (crashes the create flow)
Module	Categories
Discovered by	Test #9 — "Create category (valid)"
Symptom	POST /api/categories with body {name, color} returned HTTP 500. Backend log: Error: set failed: value argument contains undefined in property 'categories.<id>.icon'
Root cause	In backend/src/categories/categories.service.ts , the create() method did <code>const category = { id, ...dto, color: dto.color ?? '#6366f1', userId, ... }</code> . After class-validator transform, <code>dto.icon</code> was present-but-

	undefined. The spread carried <code>icon: undefined</code> into the Firebase <code>.set()</code> call. The Firebase Admin SDK rejects undefined values explicitly.
Fix	Replaced the spread with a <code>for...of Object.entries(dto)</code> loop that copies only defined fields. Applied the same fix to <code>update()</code> since it had the identical pattern.
Preventive fixes elsewhere	Same vulnerable pattern existed in other services. Patched preemptively: • backend/src/tasks/tasks.service.ts — <code>update()</code> • backend/src/goals/goals.service.ts — <code>update()</code> for goals + <code>updateMilestone()</code> via new <code>definedOnly()</code> helper • backend/src/habits/habits.service.ts — <code>update()</code>
Verification	Re-ran tests → all 4 category tests + the projects regression test pass
Files changed	4 backend service files (categories, tasks, goals, habits)

BUG-002 — Auth: duplicate email returned HTTP 400 instead of 409

Field	Value
Severity	Low (cosmetic / REST convention)
Module	Auth
Discovered by	Test #2 — "Register with duplicate email returns 409"
Symptom	Re-registering an existing email returned <code>400 Bad Request</code> instead of <code>409 Conflict</code>
Root cause	backend/src/auth/auth.service.ts — <code>register()</code> threw <code>BadRequestException</code> for the "Email or username already taken" case. The REST-correct status for "resource already exists" is 409.
Fix	Imported <code>ConflictException</code> (replacing <code>BadRequestException</code>) and swapped the throw.
Verification	Test now passes with the corrected expectation
Files changed	1 backend service file

5. Test runner issues fixed (no app code changes)

During the test runner development I hit several PowerShell-specific issues that aren't app bugs but are worth recording:

#	Issue	Fix
1	Em-dashes (–) in source comments saved as UTF-8 broke PowerShell 5.1 parsing	Replaced with -
2	Emoji literals (👉, 😊, ✱) in test data broke PowerShell parsing	Replaced with plain text (good, ok, S)
3	?? null-coalescing operator (PowerShell 7+ only) failed on PS 5.1	Replaced with explicit <code>if (\$null -eq ...)</code>
4	Unescaped & in URL string (<code>year=Y&month=M</code>) interpreted as command separator	Concatenated with + '&' +
5	Helper function named <code>h</code> collided with <code>Get-History</code> alias → JWT being passed to wrong cmdlet	Renamed to <code>MakeHeaders</code>
6	Apostrophes in test names (<code>user A's task</code>) caused parser confusion	Switched to backtick (<code>user A\'s task</code>)
7	Test #8 used the original refresh token <i>after</i> a login call rotated it → false 401	Sequenced test to use the freshly-issued token

6. Untested areas (cannot reach with API tests)

These require a real browser; they were not exercised:

Area	Why untested	Suggested verification
Pomodoro timer ticking (countdown, pause/resume across navigation, ring animation)	Needs running Angular app in browser	Manual test
Web Audio beep on session completion	Requires audio playback	Manual test
Google Calendar OAuth flow (auth, callback, scope upgrade)	Needs Google consent screen	Manual test (and existing screenshots show it working)
GET /api/google/events actual data	Requires a connected Google account in the test env	Manual test
All Angular UI: forms, modals, navigation, routerLink, RouterLinkActive	Frontend rendering	Manual test
Theme toggle + localStorage persistence	Browser-only	Manual test
Focus settings persistence across page reload	localStorage-based	Manual test
Calendar view: month grid, day-cell click, weekend/birthday/office-hours overlays	DOM rendering	Manual test
Optimistic UI updates (the "feels instant" UX)	Tested protocol layer, not visual smoothness	Manual test
Concurrent edits / two-tab race conditions	Out of scope for sequential API tests	Future load testing

7. Code-review findings (latent issues, not test failures)

These didn't cause any test to fail but are real concerns surfaced by reading the code:

#	Area	Issue	Severity
CR-1	Security	No rate limiting on /auth/login — brute-forceable. Fix: @nestjs/throttler decorator.	High
CR-2	Security	Refresh tokens stored in plain text on the user record. Anyone with Firebase read access can replay them. Should be hashed.	High
CR-3	Security	No password complexity rules beyond min 8. 12345678 is accepted.	Med
CR-4	Security	No email verification flow. Users can register with any email.	Med
CR-5	Scalability	Every list endpoint uses firebase.getList('collection') which fetches all users' records , then filters by userId in JS. Fine at < 1k records; slows down at 10k+.	Med (latent)
CR-6	Scalability	/analytics/summary fetches 8 collections per call (tasks, habits, checkins, sessions, journal, projects, payments, categories) with no caching.	Low

CR-7	Correctness	Project <code>paymentStatus</code> does NOT auto-flip to <code>PAID</code> when total received reaches the quoted amount via payment add. Manual flip needed.	Low
CR-8	Correctness	Habit reminders are stored (<code>reminderEnabled: true</code>) but never delivered. No notification system exists.	Low
CR-9	Correctness	<code>taskToEvent()</code> in <code>google-calendar.service.ts</code> hardcodes <code>timeZone: 'Asia/Kolkata'</code> . Wrong for non-IST users.	Low
CR-10	Correctness	Focus sessions don't validate <code>endedAt > startedAt</code> OR <code>actualMinutes ≤ plannedMinutes + tolerance</code> . A malformed client could log a 6-hour "WORK session" from a 25-minute plan.	Low
CR-11	UX	No confirmation when navigating away from an unsaved journal entry.	Low
CR-12	UX	No way to manually mark a non-trip day as off (sick/leave). Only type: <code>'trip'</code> plans drive auto-off.	Low
CR-13	UX	No undo for delete operations. Confirm dialog is the only safety net.	Low
CR-14	UX	Some forms don't surface field-level validation errors — they just keep the submit button disabled, so the user can't tell <i>what</i> is wrong.	Low

8. Enhancement ideas (prioritized by leverage)

Tier 1 — High value, low effort

1. **Manual "day off" toggle** — extend the trip auto-detection with a `Mark today as off` button on the habits page so sick/leave days don't break streaks either. Reuses the same `offDates` plumbing.
2. **Auto-update payment status** — when total received reaches the quoted amount, auto-flip (or auto-suggest) `paymentStatus` → `PAID`.
3. **Search bar** in My Plans + Projects + Journal — a `?q=` query param on list endpoints goes a long way.
4. **Rate-limit** `/auth/login` — `@nestjs/throttler` is a one-decorator install.
5. **Hash refresh tokens** in DB. ~15 minutes of work.
6. **Confirm before discarding unsaved journal entry** when navigating away (Angular `CanDeactivate` guard).
7. **Export to JSON** — single endpoint that dumps all the user's data, useful for backup.
8. **Toast on habit completion from dashboard** — already partially there; just polish.

Tier 2 — Medium effort, real value

9. **Recurring task templates** — define once, scheduler auto-creates instances. Needs a cron worker.
10. **Smart reminders hub** — combine habit `reminderEnabled`, task `dueDate` alerts, deadline notifications. Web Notification API for v1 (in-tab); push for v2.
11. **Tags / labels** — lightweight cross-module tagging (a task, a habit, a journal entry can share tag "work").
12. **Bulk operations** — multi-select on My Plans for status / category / delete.
13. **Calendar view of journal entries** — colored mood dot per day on the existing calendar.
14. **Insights v2** — week-over-week trends, best-day-of-week for focus, etc.

Tier 3 — Bigger projects

15. **Mobile-optimized PWA** — install banner, offline mode, push notifications.
16. **Backup / restore to Google Drive** (Drive scope already implicitly available).
17. **AI helpers** — Claude API for journal prompts, weekly summary, habit recommendations.
18. **Year-in-review public share** — read-only link to share annual stats.

Tier 4 — Polish

- Loading skeletons instead of spinners

- Keyboard shortcuts (`n` = new task, `j` / `k` = navigate, `t` = today)
- Drag-and-drop reordering for tasks (only milestones currently have it)
- Empty-state illustrations
- Onboarding tour

9. Verdict

The backend API is **solid**. All happy paths, all error paths, all validation, and all cross-user isolation tests pass. The trip-day auto-exclusion that's central to the habit workflow works end-to-end. The two bugs discovered during testing have been fixed and verified, and the same vulnerable pattern (`{ ...dto }` spread → Firebase `undefined` rejection) has been removed from every other service preemptively.

Recommended next step: manual UI walkthrough of the areas listed in §6, then pick from §8 Tier 1 enhancements based on what you actually use.

10. Files modified during testing

File	Reason
backend/src/categories/categories.service.ts	Fix BUG-001 (defined-only spread on create + update)
backend/src/tasks/tasks.service.ts	Preemptive same-pattern fix in update()
backend/src/goals/goals.service.ts	Preemptive same-pattern fix in update() + updateMilestone() via new definedOnly() helper
backend/src/habits/habits.service.ts	Preemptive same-pattern fix in update()
backend/src/auth/auth.service.ts	Fix BUG-002 (ConflictException for duplicate email)
test-runner.ps1 (new)	The test suite itself — re-runnable
TEST-REPORT.md (new — this file)	The report you are reading